

Modern Reinforcement Learning: From Agents to LLMs

Learning from Preferences

Lecture 7

Keywords: Preferences · Reward modeling · Pairwise comparisons · Bradley–Terry · Learned rewards

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Lecture 7: Learning from Preferences & Reward Modeling

- Learning from Preferences
 - Reward modeling
 - Pairwise comparisons
 - Bradley–Terry models
 - Learned rewards in RL
 - From control to alignment



Side note / “Keep in mind” box:

— Rewards are not ground truth — they are models learned from data.

Why rewards are hard to specify

- Why rewards are hard
 - Proxy objectives
 - Misspecification
 - Reward hacking (<https://openai.com/index/faulty-reward-functions/>)
 - Human values $\neq$ scalar functions

Preferences as Indirect Supervision

- Preferences as supervision
 - Relative judgments
 - Easier than scoring
 - Partial information

 Side note / “Keep in mind” box:
— Preferences tell us what is better, not how good something is.

Where Preference Learning Fits in RL

- Learning objectives from data
 - Behavior data vs preference data
 - Reward learning vs policy learning
 - Offline and online settings
 - Two-stage learning pipeline

What Is Preference Feedback?

- Preference feedback
 - Comparisons between trajectories
 - $\tau_i \succ \tau_j$ indicates preferences
 - No absolute reward values
 - Possibly noisy and inconsistent

Preference Learning Problem Setup

- Preference learning setup
 - MDP without known reward
 - Trajectories $\tau = (s_0, a_0, \dots, s_T)$
 - Latent reward $r_\theta(s, a)$
 - Preference data $\mathcal{D} = \{(\tau_i \succ \tau_j)\}$



Side note / “Keep in mind” box:
— The reward is a latent variable,
preferences are the observations.

Bradley–Terry Preference Model

Preference model (Bradley–Terry):

- Trajectory return

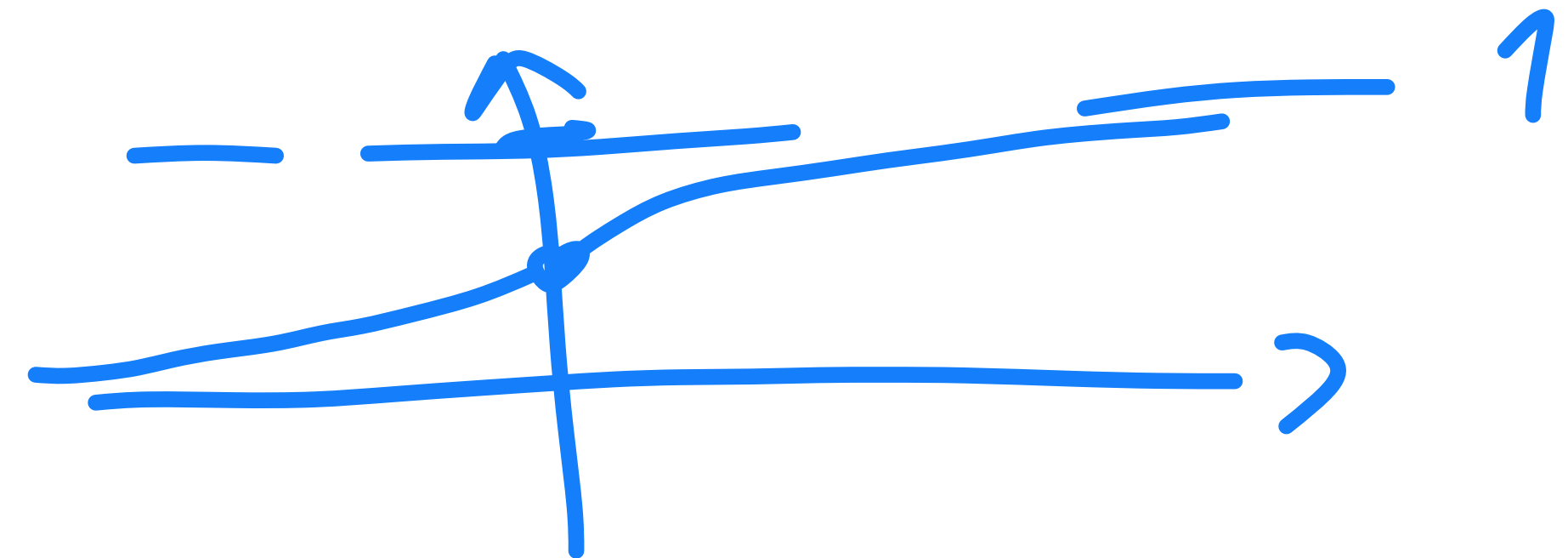
$$R_{\theta}(\tau) = \sum_{t=0}^T r_{\theta}(s_t, a_t)$$

- Pairwise preference likelihood

$$P_r(\tau_i > \tau_j | \theta) = \zeta(R_{\theta}(\tau_i) - R_{\theta}(\tau_j))$$

- Logistic noise assumption

$$\zeta(x) = \frac{1}{1 + \exp(-x)}$$



- Only differences in return are identifiable

Log-Likelihood of Preference Data

Preference learning objective

- Preference dataset

$$\mathcal{D} = \left\{ (\tau_i, \tau_j, y_{ij}) \right\}_{i,j=1}^N \quad y_{ij} = \mathbb{1} \{ \tau_i > \tau_j \} \in \{0, 1\}$$

- Probabilistic model

$$\Delta_{ij} = R_{\theta}(\tau_i) - R_{\theta}(\tau_j) \quad y_{ij} \sim \text{Bern}(b(\Delta_{ij}))$$

$$L(\theta) = \prod_{(i,j) \in \mathcal{D}} b(\Delta_{ij})^{y_{ij}} (1 - b(\Delta_{ij}))^{1-y_{ij}}$$

$$\mathcal{L}(\theta) = - \sum_{(i,j) \in \mathcal{D}} y_{ij} \log b(\Delta_{ij}) + (1 - y_{ij}) \log (1 - b(\Delta_{ij}))$$

Identifiability and Invariances

- Identifiability of reward models
- Additive invariance

$$r_{\theta}(s, a) \mapsto r_{\theta}(s, a) + c \quad \Rightarrow \quad \mathbb{P}(\tau_i \succ \tau_j) \text{ unchanged}$$

- Return differences are invariant

$$R_{\theta}(\tau_i) - R_{\theta}(\tau_j)$$

- Absolute reward scale is unidentifiable
- Multiple reward functions explain same preference

 Side note / “Keep in mind” box:
— Reward learning identifies an equivalence class of rewards, not a unique function.

Gradient and Optimization Properties

Optimization of preference objective

- Gradient of log-likelihood

$$\nabla_{\theta} \ell(\theta) = - \sum_{(i,j) \in \mathcal{D}} (y_{ij} - \sigma(\Delta R_{ij})) \nabla_{\theta} \Delta R_{ij}$$

where

$$\Delta R_{ij} := R_{\theta}(\tau_i) - R_{\theta}(\tau_j)$$

- Linear reward $\Rightarrow$ convex objective
- Neural reward $\Rightarrow$ non-convex optimization

Structure of the Learned Reward

What the reward model learns

- Return decomposition

$$R_{\theta}(\tau) = \sum_{t=0}^T r_{\theta}(s_t, a_t)$$

- Gradient contribution

$$\nabla_{\theta} \Delta R_{ij} = \sum_t \nabla_{\theta} r_{\theta}(s_t^i, a_t^i) - \sum_t \nabla_{\theta} r_{\theta}(s_t^j, a_t^j)$$

- Credit assignment across trajectories
- No supervision at single time steps

Structure of the Learned Reward

Generalization limits

- Preferences collected under policy π_{data}
- Reward evaluated under policy π_{θ}
- Off-policy reward learning
- Extrapolation beyond observed trajectories

Optimization Mismatch with Learned Rewards

Objective mismatch

- True (unknown) objective

$$J^{\star}(\pi) := \mathbb{E}_{\tau \sim \pi} [R^{\star}(\tau)]$$

- Learned reward objective

$$J_{\theta}(\pi) := \mathbb{E}_{\tau \sim \pi} [R_{\theta}(\tau)]$$

- Policy optimization step

$$\pi_{k+1} \in \arg \max_{\pi} J_{\theta}(\pi)$$

- No guarantee:

$$J^{\star}(\pi_{k+1}) \geq J^{\star}(\pi_k)$$

Error Amplification Under Policy Optimization

Error amplification

- Reward model error

$$\varepsilon(\tau) := R_{\theta}(\tau) - R^{\star}(\tau)$$

- Decomposition of learned objective

$$J_{\theta}(\pi) = J^{\star}(\pi) + \mathbb{E}_{\tau \sim \pi}[\varepsilon(\tau)]$$

- Policy improvement step

$$\pi_{k+1} \in \arg \max_{\pi} \left(J^{\star}(\pi) + \mathbb{E}_{\tau \sim \pi}[\varepsilon(\tau)] \right)$$

- Bias toward trajectories with large $\varepsilon(\tau)$

Modern Parallels: Reward Models & RLHF

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Preference learning for LLMs

- Prompt–response pairs (x, y)
- Human preference data $(x, y_i \succ y_j)$
- Reward model $R_\theta(x, y) \in \mathbb{R}$
- Trained via Bradley–Terry objective

$$\mathbb{P}(y_i \succ y_j \mid x) = \sigma(R_\theta(x, y_i) - R_\theta(x, y_j))$$

 Side note / “Keep in mind” box:
— Reward models for LLMs are trajectory-level rewards over token sequences.

RLHF as Two-Stage Optimization

Decomposition of RLHF

- Stage 1: Reward learning

Preference data: $(x, y_i \succ y_j) \Rightarrow R_\theta(x, y)$

- Stage 2: Policy optimization

Language model policy $\pi_\phi(y \mid x)$

$$\phi^\star \in \arg \max_{\phi} \mathbb{E}_{y \sim \pi_\phi(\cdot|x)} [R_\theta(x, y)]$$

Scalarization and Value Collapse

- Limits of scalar reward models
- Human preferences are multi-dimensional $v(x, y) \in \mathbb{R}^k$
- Reward model learns scalarization $R_\theta(x, y) \approx w^\top v(x, y)$
- Weights w are implicit and data-dependent
- Different preferences $\Rightarrow$ same scalar reward

 Side note / “Keep in mind” box:
— Scalar rewards hide trade-offs rather than resolving them.

Failure Modes in Practice

- Observed issues in RLHF systems
 - Over-optimization of R_θ
 - Distribution shift: $(x, y)_{\text{train}} \neq (x, y)_{\text{deploy}}$
 - Reward hacking in language space
 - Sensitivity to prompt and decoding changes

Demo

Preference Learning in a Gridworld

Setup

- Gridworld MDP $(\mathcal{S}, \mathcal{A}, P, \gamma)$, unknown $r^\star$
- Trajectories $\tau = (s_0, \dots, s_T)$
- Simulated preferences: $\mathbb{P}(\tau_i \succ \tau_j) = \sigma\left(R^\star(\tau_i) - R^\star(\tau_j)\right)$
- Learn $r_\theta(s)$ from comparisons:

$$\theta^\star = \arg \min_{\theta} - \sum_{(i,j)} \log \sigma(R_\theta(\tau_i) - R_\theta(\tau_j))$$

Reward Heatmaps and Policy Change

Outcome

- Learned reward close to true reward
- Heatmaps differ slightly
- Optimal paths differ: $\pi_{r^\star}^\star \neq \pi_{r_\theta}^\star$
- Takeaway: small reward errors $\rightarrow$ large behavioral changes